

Privacy-Aware Data Management and Information Security

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Motivation

Ubiquitous, pervasive, open systems

- new ways to establish and manage identities
- new opportunities to enhance privacy
- new needs for privacy
- new risks for privacy

Scenario

Integration and sharing of different services of different parties in an open environment

- involving **dynamic collaboration** and process integration
- involving **sharing of information** from different authorities
- involving **multiplicity** and **heterogeneity** of entities and **security specifications**
- in a possible **mobile/dynamic** environment
- parties may be **unknown**
- **lots of data** communicated, collected, shared (information explosion!)

Outline

- Data Privacy: k -anonymity
- Privacy in Data Outsourcing
- Data Fragmentation and Encryption
- Authorization Enforcement in Collaborative Distributed Scenarios

Data Privacy: k -anonymity

Data collection and disclosure

- Internet provides unprecedented opportunities for the collection and sharing of privacy-sensitive information from and about users
- Information about users is collected every day
- Users have very strong concerns about the privacy of their personal information
- Protecting privacy requires the investigation of different issues, including the problem of [protecting released information against inference and linking attacks](#) which are becoming easier and easier because of the increased information availability and ease of access

Statistical DBMS vs statistical data

Often **statistical data** (or data for statistical purpose) are released

- **statistical DBMS** [AW-89]
 - the DBMS responds only to statistical queries
 - need run time checking to control information (indirectly) released
- **statistical data** [CDFS-07b]
 - publish statistics
 - control on indirect release performed before publication

Disclosure risk

Statistical data, even if 'anonymized', can be used to infer information that was not intended for disclosure

Disclosure can:

- occur based on the released data alone
- result from combination of the released data with publicly available information
- be possible only through combination of the released data with detailed external data sources that may or may not be available to the general public

When releasing data, the **disclosure risk** of sensitive information should be very low

Macrodata vs microdata

- In the past data were mainly released in tabular form (macrodata) and through statistical databases
- Today many situations require that the specific stored data themselves, called microdata, be released
 - increased flexibility and availability of information for the users
- Microdata are subject to a greater risk of privacy breaches
- The main requirements that must be taken into account are:
 - identity disclosure protection
 - attribute disclosure protection
 - inferential disclosure protection

Macrodata

Macrodata tables can be classified into the following two groups (types of tables)

- **Count/Frequency.** Each cell of the table contains the number of respondents (count) or the percentage of respondents (frequency) that have the same value over all attributes of analysis associated with the table
- **Magnitude data.** Each cell of the table contains an aggregate value of a *quantity of interest* over all attributes of analysis associated with the table

Count table – Example

Two-dimensional table showing the number of beneficiaries by county and size of benefit

County	Benefit						Total
	\$0-19	\$20-39	\$40-59	\$60-79	\$80-99	\$100+	
A	2	4	18	20	7	1	52
B	-	-	7	9	-	-	16
C	-	6	30	15	4	-	55
D	-	-	2	-	-	-	2

Magnitude table – Example

Average number of days spent in the hospital by respondents with a disease

	Hypertension	Obesity	Chest Pain	Short Breath	Tot
M	2	8.5	23.5	3	37
F	3	30.5	0	5	38.5
Tot	5	39	23.5	8	75.5

Microdata table – Example

Records about delinquent children in county Alfa

N	Child	County	Educ. HH	Salary HH	Race HH
1	John	Alfa	very high	201	black
2	Jim	Alfa	high	103	white
3	Sue	Alfa	high	77	black
4	Pete	Alfa	high	61	white
5	Ramesh	Alfa	medium	72	white
6	Dante	Alfa	low	103	white
7	Virgil	Alfa	low	91	black
8	Wanda	Alfa	low	84	white
9	Stan	Alfa	low	75	white
10	Irmis	Alfa	low	62	black
11	Renee	Alfa	low	58	white
12	Virginia	Alfa	low	56	black
13	Mary	Alfa	low	54	black
14	Kim	Alfa	low	52	white
15	Tom	Alfa	low	55	black
16	Ken	Alfa	low	48	white
17	Mike	Alfa	low	48	white
18	Joe	Alfa	low	41	black
19	Jeff	Alfa	low	44	black
20	Nancy	Alfa	low	37	white

Information disclosure

Disclosure relates to attribution of sensitive information to a respondent (an individual or organization)

There is disclosure when:

- a respondent is identified from released data (**identity disclosure**)
- sensitive information about a respondent is revealed through the released data (**attribute disclosure**)
- the released data make it possible to determine the value of some characteristic of a respondent even if no released record refers to the respondent (**inferential disclosure**)

Identity disclosure

It occurs if a third party can identify a respondent from the released data

Revealing that an individual is a respondent in a data collection may or may not violate confidentiality requirements

- Macrodata: revealing identity is generally not a problem, unless the identification leads to divulging confidential information (attribute disclosure)
- Microdata: identification is generally regarded as a problem, since microdata records are detailed; identity disclosure usually implies in this case also attribute disclosure

Attribute disclosure

It occurs when confidential information about a respondent is revealed and can be attributed to it; confidential information may be:

- revealed exactly
- closely estimated

Inferential disclosure

It occurs when information can be inferred with high confidence from statistical properties of the released data

E.g., the data may show a high correlation between income and purchase price of home. As purchase price of home is typically public information, a third party might use this information to infer the income of a respondent

Inference disclosure not always represents a risk:

- statistical data are released for enabling users to infer and understand relationships between variables
- inferences are designed to predict aggregate behavior, not individual attributes, and are then often poor predictors of individual data values

Restricted data and restricted access (1)

The choice of statistical disclosure limitation methods depends on the nature of the data products whose confidentiality must be protected

Some microdata include explicit identifiers (e.g., name, address, or Social Security number)

Removing such identifiers is a first step in preparing for the release of microdata for which the confidentiality of individual information must be protected

Restricted data and restricted access (2)

Confidentiality can be protected by:

- restricting the **amount of information** in the released tables (restricted data)
- imposing **conditions on access** to the data products (restricted access)
- some combination of these two strategies

Disclosure protection techniques

The protection techniques include:

- **sampling**: data confidentiality is protected by conducting a sample survey rather than a census
- **special rules**: designed for specific tables, they impose restrictions on the level of detail that can be provided in a table
- **threshold rule**: rules that protect sensitive cells, e.g.,
 - cell suppression
 - random rounding
 - controlled rounding
 - confidentiality edit

The anonymity problem

- The amount of privately owned records that describe each citizen's finances, interests, and demographics is increasing every day
- These data are **de-identified** before release, that is, any explicit identifier (e.g., SSN) is removed
- De-identification is not sufficient
- Most municipalities sell population registers that include the identities of individuals along with basic demographics
- These data can then be used for linking identities with de-identified information $\Rightarrow$ **re-identification**

The anonymity problem – Example

SSN	Name	Race	Date of birth	Sex	ZIP	Marital status	Disease
		asian	64/04/12	F	94142	divorced	hypertension
		asian	64/09/13	F	94141	divorced	obesity
		asian	64/04/15	F	94139	married	chest pain
		asian	63/03/13	M	94139	married	obesity
		asian	63/03/18	M	94139	married	short breath
		black	64/09/27	F	94138	single	short breath
		black	64/09/27	F	94139	single	obesity
		white	64/09/27	F	94139	single	chest pain
		white	64/09/27	F	94141	widow	short breath

Name	Address	City	ZIP	DOB	Sex	Status
.....						
.....						
Sue J. Doe	900 Market St.	San Francisco	94142	64/04/12	F	divorced
.....						

Classification of attributes in a microdata table

The attributes in the original microdata table can be classified as:

- **identifiers**. Attributes that uniquely identify a microdata respondent (e.g., SSN uniquely identifies the person with which is associated)
- **quasi-identifiers**. Attributes that, in combination, can be linked with external information to reidentify all or some of the respondents to whom information refers or reduce the uncertainty over their identities (e.g., DoB, ZIP, and Sex)
- **confidential**. Attributes of the microdata table that contain sensitive information (e.g., Disease)
- **non confidential**. Attributes that the respondents do not consider sensitive and whose release do not cause disclosure

Re-identification

A study performed in 2000 reported that the US population was uniquely identifiable by:

- year of birth, 5-digit ZIP code: 0,2%
- year of birth, county: 0,0%
- year and month of birth, 5-digit ZIP code: 4,2%
- year and month of birth, county: 0,2%
- year, month, and day of birth, 5-digit ZIP code: 63,3%
- year, month, and day of birth, county: 14,8%

Factors contributing to disclosure risk (1)

Possible sources of the disclosure risk of microdata

- **Existence of high visibility records.** Some records on the file may represent respondents with unique characteristics such as very unusual jobs (e.g., movie star) or very large incomes
- **Possibility of matching the microdata with external information.** There may be individuals in the population who possess a unique or peculiar combination of the characteristic variables on the microdata
 - if some of those individuals happen to be chosen in the sample of the population, there is a disclosure risk
 - note that the identity of the individuals that have been chosen should be kept secret

Factors contributing to disclosure risk (2)

The possibility of linking or its precision increases with:

- the existence of a high number of common attributes between the microdata table and the external sources
- the accuracy or resolution of the data
- the number of outside sources, not all of which may be known to the agency releasing the microdata

Factors contributing to decrease the disclosure risk (1)

- A microdata table often contains a subset of the whole population
 - this implies that the information of a specific respondent, which a malicious user may want to know, may not be included in the microdata table
- The information specified in microdata tables released to the public is not always up-to-date (often at least one or two-year old)
 - the values of the attributes of the corresponding respondents may have been changed in the meanwhile
 - the age of the external sources of information used for linking may be different from the age of the information contained in the microdata table

Factors contributing to decrease the disclosure risk (2)

- A microdata table and the external sources of information naturally contain noise that decreases the ability to link the information
- A microdata table and the external sources of information can contain data expressed in different forms thus decreasing the ability to link information

Measures of risk

Measuring the disclosure risk requires considering

- the probability that the respondent for whom an intruder is looking for is represented on both the microdata and some external file
- the probability that the matching variables are recorded in a linkable way on the microdata and on the external file
- the probability that the respondent for whom the intruder is looking for is unique (or peculiar) in the population of the external file

The percentage of records representing respondents who are unique in the population (**population unique**) plays a major role in the disclosure risk of microdata (with respect to the specific respondent)

Note that each population unique is a sample unique; the vice-versa is not true

k -anonymity [S-01] (1)

- k -anonymity, together with its enforcement via **generalization** and **suppression**, has been proposed as an approach to protect respondents' identities while releasing truthful information
- k -anonymity tries to capture the following requirement:
 - the released data should be indistinguishably related to no less than a certain number of respondents
- **Quasi-identifier**: Set of attributes that can be exploited for linking (whose release must be controlled)

k -anonymity (2)

- Basic idea: translate the k -anonymity requirement on the released data
 - each release of data must be such that every combination of values of quasi-identifiers can be indistinctly matched to at least k respondents
- In the released table the respondents must be indistinguishable (within a given set) with respect to a set of attributes
- k -anonymity requires that each quasi-identifier value appearing in the released table must have at least k occurrences
 - sufficient condition for the satisfaction of k -anonymity requirement

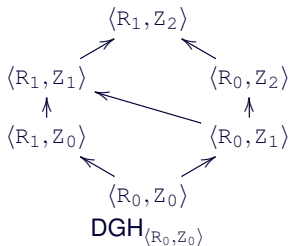
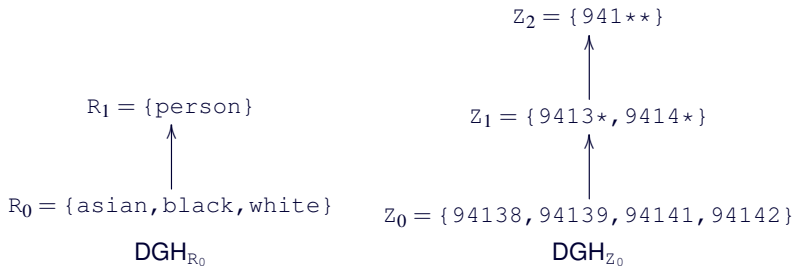
Generalization and suppression

- **Generalization.** The values of a given attribute are substituted by using more general values. Based on the definition of a generalization hierarchy
 - Example: consider attribute ZIP code and suppose that a step in the corresponding generalization hierarchy consists in suppressing the least significant digit in the ZIP code
With one generalization step: 20222 and 20223 become 2022*; 20238 and 20239 become 2023*
- **Suppression.** It is a well-known technique that consists in protecting sensitive information by removing it
 - the introduction of suppression can reduce the amount of generalization necessary to satisfy the k -anonymity constraint

Domain generalization hierarchy

- A **generalization relationship** $\leq_D$ defines a mapping between domain D and its generalizations
- Given two domains $D_i, D_j \in \text{Dom}$, $D_i \leq_D D_j$ states that the values in domain D_j are generalizations of values in D_i
- $\leq_D$ implies the existence, for each domain D , of a **domain generalization hierarchy** $\text{DGH}_D = (\text{Dom}, \leq_D)$:
 - $\forall D_i, D_j, D_z \in \text{Dom}$:
 $D_i \leq_D D_j, D_i \leq_D D_z \implies D_j \leq_D D_z \vee D_z \leq_D D_j$
 - all maximal elements of Dom are singleton
- Given a domain tuple $DT = \langle D_1, \dots, D_n \rangle$ such that $D_i \in \text{Dom}$, $i = 1, \dots, n$, the domain generalization hierarchy of DT is $\text{DGH}_{DT} = \text{DGH}_{D_1} \times \dots \times \text{DGH}_{D_n}$

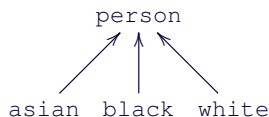
Examples of domain generalization hierarchies



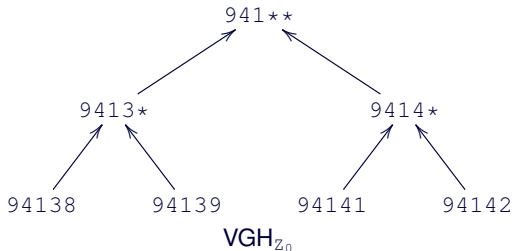
Value generalization hierarchy

- A value generalization relationship $\leq_v$ associates with each value in domain D_i a unique value in domain D_j , direct generalization of D_i
- $\leq_v$ implies the existence, for each domain D , of a value generalization hierarchy VGH_D
- VGH_D is a tree
 - the leaves are the values in D
 - the root (i.e., the most general value) is the value in the maximum element in VGH_D

Examples of value generalization hierarchies



VGH_{R_0}



VGH_{Z_0}

Generalized table with suppression

Let T_i and T_j be two tables defined on the same set of attributes. Table T_j is said to be a **generalization (with tuple suppression)** of table T_i , denoted $T_i \preceq T_j$, if:

1. $|T_j| \leq |T_i|$;
2. the domain $dom(A, T_j)$ of each attribute A in T_j is equal to, or a generalization of, the domain $dom(A, T_i)$ of attribute A in T_i ;
3. it is possible to define an injective function associating each tuple t_j in T_j with a tuple t_i in T_i , such that the value of each attribute in t_j is equal to, or a generalization of, the value of the corresponding attribute in t_i .

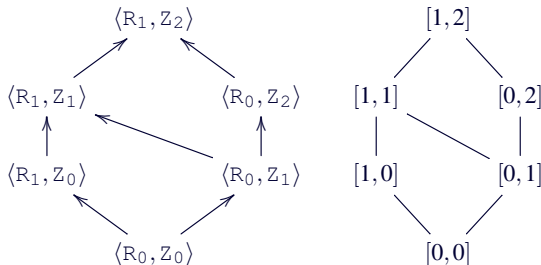
Generalized table with suppression – Example

Race	ZIP
asian	94142
asian	94141
asian	94139
asian	94139
asian	94139
black	94138
black	94139
white	94139
white	94141
PT	

Race	ZIP
person	94141
person	94139
person	94139
person	94139
person	94139
person	94139
person	94139
person	94141
GT	

k -minimal generalization with suppression (1)

- Distance vector.** Let $T_i(A_1, \dots, A_n)$ and $T_j(A_1, \dots, A_n)$ be two tables such that $T_i \preceq T_j$. The distance vector of T_j from T_i is the vector $DV_{i,j} = [d_1, \dots, d_n]$, where each d_z , $z = 1, \dots, n$, is the length of the **unique** path between $\text{dom}(A_z, T_i)$ and $\text{dom}(A_z, T_j)$ in the domain generalization hierarchy DGH_{D_z} .



k -minimal generalization with suppression (2)

Let T_i and T_j be two tables such that $T_i \preceq T_j$, and let MaxSup be the specified threshold of acceptable suppression. T_j is said to be a k -minimal generalization of table T_i iff:

1. T_j satisfies k -anonymity enforcing minimal required suppression, that is, T_j satisfies k -anonymity and $\forall T_z : T_i \preceq T_z, DV_{i,z} = DV_{i,j}, T_z$ satisfies k -anonymity $\implies |T_j| \geq |T_z|$
2. $|T_i| - |T_j| \leq \text{MaxSup}$
3. $\forall T_z : T_i \preceq T_z$ and T_z satisfies conditions 1 and 2 $\implies \neg(DV_{i,z} < DV_{i,j})$

Examples of 2-minimal generalizations

MaxSup=2

Race: R_0	ZIP: Z_0
asian	94142
asian	94141
asian	94139
asian	94139
asian	94139
black	94138
black	94139
white	94139
white	94141
PT	

Race: R_1	ZIP: Z_0
person	94141
person	94139
person	94139
person	94139
person	94139
person	94139
person	94139
person	94141
GT _[1,0]	

Race: R_0	ZIP: Z_1
asian	9414*
asian	9414*
asian	9413*
asian	9413*
asian	9413*
black	9413*
black	9413*
GT _[0,1]	

Computing a preferred generalization

Different **preference criteria** can be applied in choosing a preferred minimal generalization, among which:

- **minimum absolute distance** prefers the generalization(s) with the smallest absolute distance, that is, with the smallest total number of generalization steps (regardless of the hierarchies on which they have been taken)
- **minimum relative distance** prefers the generalization(s) with the smallest relative distance, that is, that minimizes the total number of relative steps (a step is made relative by dividing it over the height of the domain hierarchy to which it refers)
- **maximum distribution** prefers the generalization(s) with the greatest number of distinct tuples
- **minimum suppression** prefers the generalization(s) that suppresses less tuples, that is, the one with the greatest cardinality

Classification of k -anonymity techniques (1)

Generalization and suppression can be applied at different levels of granularity

- **Generalization** can be applied at the level of single column (i.e., a generalization step generalizes all the values in the column) or single cell (i.e., for a specific column, the table may contain values at different generalization levels)
- **Suppression** can be applied at the level of row (i.e., a suppression operation removes a whole tuple), column (i.e., a suppression operation obscures all the values of a column), or single cells (i.e., a k -anonymized table may wipe out only certain cells of a given tuple/attribute)

Classification of k -anonymity techniques (2)

Generalization	Suppression			
	<i>Tuple</i>	<i>Attribute</i>	<i>Cell</i>	<i>None</i>
<i>Attribute</i>	AG_TS	AG_AS $\equiv$ AG_	AG_CS	AG_ $\equiv$ AG_AS
<i>Cell</i>	CG_TS not applicable	CG_AS not applicable	CG_CS $\equiv$ CG_	CG_ $\equiv$ CG_CS
<i>None</i>	_TS	_AS	_CS	_ not interesting

2-anonymized tables wrt different models (1)

Race	DOB	Sex	ZIP
asian	64/04/12	F	94142
asian	64/09/13	F	94141
asian	64/04/15	F	94139
asian	63/03/13	M	94139
asian	63/03/18	M	94139
black	64/09/27	F	94138
black	64/09/27	F	94139
white	64/09/27	F	94139
white	64/09/27	F	94141

PT

Race	DOB	Sex	ZIP
asian	64/04	F	941**
asian	64/04	F	941**
asian	63/03	M	941**
asian	63/03	M	941**
black	64/09	F	941**
black	64/09	F	941**
white	64/09	F	941**
white	64/09	F	941**

AG_TS

2-anonymized tables wrt different models (2)

Race	DOB	Sex	ZIP
asian	*	F	*
asian	*	F	*
asian	*	F	*
asian	63/03	M	9413*
asian	63/03	M	9413*
black	64/09	F	9413*
black	64/09	F	9413*
white	64/09	F	*
white	64/09	F	*

AG_CS

Race	DOB	Sex	ZIP
asian	64	F	941**
asian	64	F	941**
asian	64	F	941**
asian	63	M	941**
asian	63	M	941**
black	64	F	941**
black	64	F	941**
white	64	F	941**
white	64	F	941**

AG_ $\equiv$ AG_AS

2-anonymized tables wrt different models (3)

Race	DOB	Sex	ZIP
asian	64	F	941**
asian	64	F	941**
asian	64	F	941**
asian	63/03	M	94139
asian	63/03	M	94139
black	64/09/27	F	9413*
black	64/09/27	F	9413*
white	64/09/27	F	941**
white	64/09/27	F	941**

CG_≡CG_{CS}

Race	DOB	Sex	ZIP

_TS

2-anonymized tables wrt different models (4)

Race	DOB	Sex	ZIP
asian	*	F	*
asian	*	F	*
asian	*	F	*
asian	*	M	*
asian	*	M	*
black	*	F	*
black	*	F	*
white	*	F	*
white	*	F	*

_AS

Race	DOB	Sex	ZIP
asian	*	F	*
asian	*	F	*
asian	*	F	*
asian	*	M	94139
asian	*	M	94139
*	64/09/27	F	*
*	64/09/27	F	94139
*	64/09/27	F	94139
*	64/09/27	F	*

_CS

Algorithms for computing a k -anonymous table

- The problem of finding minimal k -anonymous tables, with attribute generalization and tuple suppression, is computationally hard
- The majority of the exact algorithms proposed in literature have computational time exponential in the number of the attributes composing the quasi-identifier
 - when the number $|QI|$ of attributes in the quasi-identifier is small compared with the number n of tuples in the private table PT, these exact algorithms with attribute generalization and tuple suppression are practical
- Recently many exact algorithms for producing k -anonymous tables through attribute generalization and tuple suppression have been proposed

Algorithms for **AG_TS** and **AG_**

Samarati's algorithm [S-01] (1)

- Each path in DGH_{DT} represents a generalization strategy for PT
- We call **locally minimal generalization** the lowest node of each path satisfying k -anonymity
- Properties exploited by the algorithm:
 1. each k -minimal generalization is locally minimal with respect to a path (but the converse is not true)
 2. going up in the hierarchy the number of tuples that must be removed to guarantee k -anonymity decreases
- If there is no solution that guarantees k -anonymity suppressing less than MaxSup tuples at height h , there cannot exist a solution, with height lower than h that guarantees it

Samarati's algorithm (2)

- The algorithm adopts a *binary search* on the lattice of distance vectors:
 1. evaluate solutions at height $\lfloor h/2 \rfloor$
 2. if there exists at least a solution satisfying k -anonymity
 - then evaluates solutions at height $\lfloor h/4 \rfloor$
 - otherwise evaluates solutions at height $\lfloor 3h/4 \rfloor$
 3. until the algorithm reaches the lowest height for which there is a distance vector that satisfies k -anonymity
- To reduce the computational cost, it adopts a **distance vector matrix** that avoids the explicit computation of each generalized table

Samarati's algorithm – Example (1)

Distance vector matrix

Race: R_0	ZIP: Z_0
asian	94142
asian	94141
asian	94139
asian	94139
asian	94139
black	94138
black	94139
white	94139
white	94141

	t_1	t_2	$t_3/t_4/t_5$	t_6	t_7	t_8	t_9
t_1	[0,0]	[0,1]	[0,2]	[1,2]	[1,2]	[1,2]	[1,1]
t_2	[0,1]	[0,0]	[0,2]	[1,2]	[1,2]	[1,2]	[1,0]
t_6	[1,2]	[1,2]	[1,1]	[0,0]	[0,1]	[1,1]	[1,2]
t_7	[1,2]	[1,2]	[1,0]	[0,1]	[0,0]	[1,0]	[1,2]
t_8	[1,2]	[1,2]	[1,0]	[1,1]	[1,0]	[0,0]	[0,2]
t_9	[1,1]	[1,0]	[1,2]	[1,2]	[1,2]	[0,2]	[0,0]

Samarati's algorithm – Example (2)

Compute solutions at height 0: $GT_{[0,0]}$

Race: R_0	ZIP: Z_0
asian	94142
asian	94141
asian	94139
asian	94139
asian	94139
black	94138
black	94139
white	94139
white	94141

The generalized table does not satisfy 2-anonymity

Samarati's algorithm – Example (3)

Suppose $k = 2$ and $\text{MaxSup}=2$.

Compute first solutions at height 1: $\text{GT}_{[1,0]}$ and $\text{GT}_{[0,1]}$

Race: R_1	ZIP: Z_0	Race: R_0	ZIP: Z_1
<i>person</i>	94142	asian	9414*
person	94141	asian	9414*
person	94139	asian	9413*
person	94139	asian	9413*
person	94139	asian	9413*
<i>person</i>	94138	black	9413*
person	94139	black	9413*
person	94139	white	9413*
person	94141	white	9414*

Both the generalized tables satisfy 2-anonymity

k -Optimize algorithm (1) [BA-05]

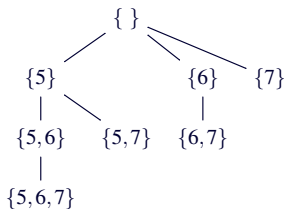
- Order attributes in Q and the values in their domains
- Associate an integer *index* value with each domain value, following the defined order

Race			ZIP			
⟨[asian]	[black]	[white]⟩	⟨[94138]	[94139]	[94141]	[94142]⟩
1	2	3	4	5	6	7

- A generalization is the union of individual index values
- The least value in an attribute domain is omitted. E.g., {6} corresponds to:
 - Race: {1}, that is: ⟨[asian or black or white]⟩
 - ZIP: {4, 6}, that is: ⟨[94138 or 94139],[94141 or 94142]⟩
- Order of values within domains has impact on generalization

k -Optimize algorithm (2)

- k -Optimize builds a **set enumeration tree** over the set I of indexes



- The root node of the tree is the empty set
- The children of n are the sets obtained by appending a single element i of I to n , such that $\forall i' \in n, i > i'$
- Each node is associated with a **cost** that reflects both the amount of generalization and suppression of the anonymization represented by the node
 $\implies$ each tuple is associated with a cost that reflects the information loss associated with its generalization or suppression

k -Optimize algorithm (3)

- k -Optimize visits the tree (e.g., using a depth-first search) for searching the anonymization with lowest cost
- Since the number of nodes in the tree is $2^{|I|}$ the visit of the tree is not practical $\Rightarrow$ pruning strategy to reduce computational cost
- Node n is pruned iff none of its descendants could be optimal
- This determination can be made by computing a lower bound on the cost of the nodes in the subtree rooted at n
 $\Rightarrow$ if the lower bound is greater than the current best cost, node n is pruned

Incognito algorithm [LDR-05]

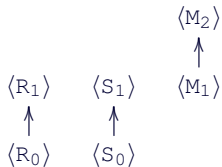
k -anonymity with respect to a proper subset of QI is a necessary (not sufficient) condition for k -anonymity with respect to QI

- **Iteration 1**: check k -anonymity for each attribute in QI , discarding generalizations that do not satisfy k -anonymity
- **Iteration 2**: combine the remaining generalizations in pairs and check k -anonymity for each couple obtained
- ...
- **Iteration i** : consider all the i -uples of attributes, obtained combining generalizations that satisfied k -anonymity at iteration $i - 1$. Discard non k -anonymous solutions
- ...
- **Iteration $|QI|$** returns the final result

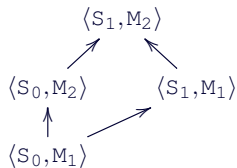
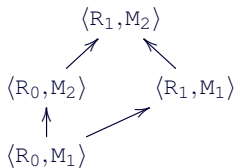
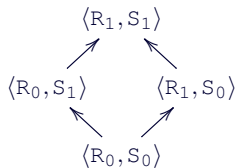
Incognito adopts a **bottom-up** approach for the visit of DGHs

Incognito – Example (1)

Iteration 1

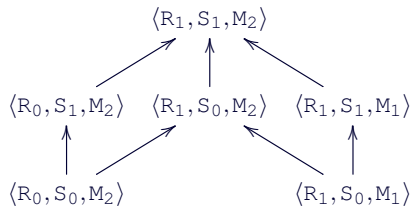


Iteration 2



Incognito – Example (2)

Iteration 3



Heuristic algorithms

- The exact algorithms have complexity exponential in the size of QI
- Heuristic algorithms have been proposed
 - [I-02]: based on genetic algorithms, it solves the k -anonymity problem using an incomplete stochastic search method
 - [W-04]: based on simulated annealing for finding locally minimal solutions, it requires high computational time and does not assure the quality of the solution
 - [FWY-05]: top-down heuristic to make a table to be released k -anonymous; it starts from the most general solution, and iteratively specializes some values of the current solution until the k -anonymity requirement is violated
- No bounds on efficiency and goodness of the solutions can be given
- Experimental results can be used to assess the quality of the solution retrieved

Algorithms for **_CS** and **CG_**

Mondrian multidimensional algorithm [LDR-06] (1)

- Each attribute in QI represents a dimension
- Each tuple in PT represents a point in the space defined by QI
- Tuples with the same QI value are represented by giving a multiplicity value to points
- The multi-dimensional space is partitioned by splitting dimensions such that each area contains at least k occurrences of point values
- All the points in a region are generalized to a unique value
- The corresponding tuples are substituted by the computed generalization

Mondrian multidimensional algorithm (2)

Mondrian algorithm is flexible and can operate

- on a different **number of attributes**
 - single-dimension
 - multi-dimension
- with different **recoding** (generalization) strategies
 - global recoding
 - local recoding
- with different **partitioning** strategies
 - strict (i.e., non-overlapping) partitioning
 - relaxed (i.e., potentially overlapping) partitioning
- using different **metrics** to determine how to split on each dimension

Mondrian multidimensional algorithm – Example (1)

Private table

Marital status	ZIP
divorced	94142
divorced	94141
married	94139
married	94139
married	94139
single	94138
single	94139
single	94139
widow	94141

widow			1	
divorced			1	1
married		3		
single	1	2		
	94138	94139	94141	94142

Mondrian multidimensional algorithm – Example (2)

3-anonymous table

Marital status	ZIP
divorced or widow	9414*
divorced or widow	9414*
married	94139
married	94139
married	94139
single	9413*
single	9413*
single	9413*
divorced or widow	9414*

widow			1	
divorced			1	1
married		3		
single	1	2		
	94138	94139	94141	94142

Approximation algorithms

- Approximation algorithms for general and specific values of k (e.g., 1.5-approximation for 2-anonymity, and 2-approximation for 3-anonymity [AFKMPTZ-05b])
- Approximation algorithm for **_CS**
 - [MW-04]: $O(k \log(k))$ -approximation
 - [AFKMPTZ-05a]: with unbounded value of k , $O(k)$ -approximation solution
- Approximation algorithm for **CG_**
 - [AFKMPTZ-05b]: with unbounded value of k , $O(k)$ -approximation solution

k -anonymity revisited [GMT-08]

- k -anonymity requirement: Each release of data must be such that every combination of values of quasi-identifiers can be indistinctly matched to at least k respondents
- When generalization is performed at attribute level (**AG**) this is equivalent to require each quasi-identifier n -uple to have at least k occurrences
- When generalization is performed at cell level (**CG**) the existence of at least k occurrences is a sufficient but not necessary condition; a less stricter requirement would suffice
 1. For each sequence of values pt in $PT[q]$ there are at least k tuples in $T[q]$ that contain a sequence of values generalizing pt
 2. For each sequence of values t in $T[q]$ there are at least k tuples in $PT[q]$ that contain a sequence of values for which t is a generalization

k -anonymity revisited – Example

Race	ZIP
white	94138
black	94139
asian	94141
asian	94141
asian	94142

PT

Race	ZIP
person	9413*
person	9413*
asian	9414*
asian	9414*
asian	9414*

2-anonymity

Race	ZIP
person	9413*
person	9413*
asian	94141
asian	9414*
asian	9414*

2-anonymity (revisited)

Race	ZIP
person	9413*
person	9413*
asian	9414*
asian	9414*
asian	94142

no 2-anonymity

Race	ZIP
person	9413*
person	9413*
asian	94141
asian	94141
asian	9414*

Attribute Disclosure

2-anonymous table according to the **AG_** model

k -anonymity is vulnerable to some attacks [MGK-06,S-01]

Race	DOB	Sex	ZIP	Disease
asian	64	F	941**	hypertension
asian	64	F	941**	obesity
asian	64	F	941**	chest pain
asian	63	M	941**	obesity
asian	63	M	941**	obesity
black	64	F	941**	short breath
black	64	F	941**	short breath
white	64	F	941**	chest pain
white	64	F	941**	short breath

Homogeneity of the sensitive attribute values

- All tuples with a quasi-identifier value in a k -anonymous table may have the same sensitive attribute value
 - an adversary knows that Carol is a black female and that her data are in the microdata table
 - the adversary can infer that Carol suffers from short breath

Race	DOB	Sex	ZIP	Disease
...	...	...	...	...
black	64	F	941**	short breath
black	64	F	941**	short breath
...	...	...	...	...

Background knowledge

- Based on prior knowledge of some additional external information
 - an adversary knows that **Hellen** is a **white female** and she is in the microdata table
 - the adversary can infer that the disease of **Hellen** is either **chest pain** or **short breath**
 - the adversary knows that the **Hellen** runs 2 hours a day and therefore that **Hellen** cannot suffer from **short breath**
⇒ the adversary infers that **Hellen's** disease is **chest pain**

Race	DOB	Sex	ZIP	Disease
...	...	...	...	...
white	64	F	941**	chest pain
white	64	F	941**	short breath

ℓ -diversity (1)

- A q -block (i.e., set of tuples with the same value for QI) in T is ℓ -diverse if it contains at least ℓ different “well-represented” values for the sensitive attribute in T
 - “well-represented” different definitions based on entropy or recursion (e.g., a q -block is ℓ -diverse if removing a sensitive value it remains $(\ell-1)$ -diverse)
- ℓ -diversity: an adversary needs to eliminate at least $\ell-1$ possible values to infer that a respondent has a given value

ℓ -diversity (2)

- T is ℓ -diverse if all its q -blocks are ℓ -diverse
 - $\implies$ the homogeneity attack is not possible anymore
 - $\implies$ the background knowledge attack becomes more difficult
- ℓ -diversity is monotonic with respect to the generalization hierarchies considered for k -anonymity purposes
- Any algorithm for k -anonymity can be extended to enforce the ℓ -diverse property

Skewness attack

ℓ -diversity leaves space to attacks based on the distribution of values inside q -blocks

- **Skewness attack** occurs when the distribution in a q -block is different than in the original population
- 20% populations suffers from diabetes; 75% of tuples in a q -block have diabetes
⇒ people in the q -block have higher probability of suffering from diabetes

Race	DOB	Sex	ZIP	Disease
black	64	F	941**	diabetes
black	64	F	941**	short breath
black	64	F	941**	diabetes
black	64	F	941**	diabetes

Similarity attack

- **Similarity attack** happens when a q -block has different but semantically similar values for the sensitive attribute

Race	DOB	Sex	ZIP	Disease
black	64	F	941**	stomach ulcer
black	64	F	941**	stomach ulcer
black	64	F	941**	gastritis

Group closeness [LLV-07]

- A q -block respects t -closeness if the distance between the distribution of the values of the sensitive attribute in the q -block and in the considered population is lower than t
- T respects t -closeness if all its q -blocks respect t -closeness
- t -closeness is monotonic with respect to the generalization hierarchies considered for k -anonymity purposes
- Any algorithm for k -anonymity can be extended to enforce the t -closeness property, which however might be difficult to achieve

External knowledge [CLR-07,MKMGH-07] (1)

- The consideration of the **adversary's background knowledge** (or **external knowledge**) is necessary when reasoning about privacy in data publishing
- External knowledge can be exploited for inferring sensitive information about individuals with high confidence
- Positive inference
 - a respondent **has** a given value (or a value within a restricted set)
- Negative inference
 - a respondent **does not have** a given value
- Existing approaches have mostly focused on positive inference

External knowledge (2)

- External knowledge may include:
 - similar datasets released by different organizations
 - instance-level information
 - ...
- Not possible to know a-priori what external knowledge the adversary possesses
- It is necessary to provide the data owner with a means to specify adversarial knowledge

External knowledge modeling [CLR-07]

- An adversary has knowledge about an individual (target) represented in a released table and knows the individual's QI values
⇒ **goal**: predict whether the target has a target sensitive value
- External knowledge modeled through a logical expression
- Three basic classes of expressions, representing knowledge about:
 - **the target individual**. Information that the adversary may know about the target individual
 - **others**. Information about individuals other than the target
 - **same-value families**. The knowledge that a group (or family) of individuals have the same sensitive value
- Other types of external knowledge may be identified.....

External knowledge – Example (1)

Name	DOB	Sex	ZIP	Disease
Alice	74/04/12	F	94142	aids
Bob	74/04/13	F	94141	flu
Carol	74/09/15	F	94139	flu
David	74/03/13	M	94139	aids
Elen	64/03/18	M	94139	flu
Frank	64/09/27	F	94138	short breath
George	64/09/27	F	94139	flu
Harry	64/09/27	M	94139	aids

Original table



DOB	Sex	ZIP	Disease
74	*	941**	aids
74	*	941**	flu
74	*	941**	flu
74	*	941**	aids
64	*	941**	flu
64	*	941**	short breath
64	*	941**	flu
64	*	941**	aids

4-anonymized table

Released table is 4-anonymized but

External knowledge – Example (2)

DOB	Sex	ZIP	Disease
74	*	941**	aids
74	*	941**	flu
74	*	941**	flu
74	*	941**	aids
64	*	941**	flu
64	*	941**	short breath
64	*	941**	flu
64	*	941**	aids

4-anonymized table

An adversary knows that Harry, born in 64 and living in area 94139, is in the table

External knowledge – Example (2)

DOB	Sex	ZIP	Disease
74	*	941**	aids
74	*	941**	flu
74	*	941**	flu
74	*	941**	aids
64	*	941**	flu
64	*	941**	short breath
64	*	941**	flu
64	*	941**	aids

4-anonymized table



DOB	Sex	ZIP	Disease
64	*	941**	flu
64	*	941**	short breath
64	*	941**	flu
64	*	941**	aids

4-anonymized table

An adversary knows that Harry, born in 64 and living in area 94139, is in the table

⇒ Harry belongs to the second group

⇒ Harry has aids with confidence 1/4

External knowledge – Example (3)

DOB	Sex	ZIP	Disease
-----	-----	-----	---------

64	*	941**	flu
64	*	941**	short breath
64	*	941**	flu
64	*	941**	aids

4-anonymized table

From another dataset, the adversary knows that George (who is in the table, is born in 64, and lives in area 941**) has flu

External knowledge – Example (3)

DOB	Sex	ZIP	Disease
-----	-----	-----	---------

64	*	941**	flu
64	*	941**	short breath
64	*	941**	flu
64	*	941**	aids

4-anonymized table



DOB	Sex	ZIP	Disease
-----	-----	-----	---------

64	*	941**	short breath
64	*	941**	flu
64	*	941**	aids

4-anonymized table

From another dataset, the adversary knows that George (who is in the table, is born in 64, and leaves in area 941**) has flu

⇒ Harry has aids with confidence 1/3

External knowledge – Example (4)

DOB	Sex	ZIP	Disease
-----	-----	-----	---------

64	*	941**	short breath
64	*	941**	flu
64	*	941**	aids

4-anonymized table

From personal knowledge, the adversary knows that Harry does not have short breath

External knowledge – Example (4)

DOB	Sex	ZIP	Disease
-----	-----	-----	---------

DOB	Sex	ZIP	Disease
-----	-----	-----	---------



64	*	941**	short breath
64	*	941**	flu
64	*	941**	aids

4-anonymized table

64	*	941**	flu
64	*	941**	aids

4-anonymized table

From personal knowledge, the adversary knows that Harry does not have short breath

⇒ Harry has aids with confidence 1/2

External knowledge – Example (5)

DOB	Sex	ZIP	Disease
74	*	941**	aids
74	*	941**	flu
74	*	941**	flu
74	*	941**	aids
64	*	941**	flu
64	*	941**	short breath
64	*	941**	flu
64	*	941**	aids

4-anonymized table

An adversary knows that Bob, born in 74 and living in area 94141, is in the table

External knowledge – Example (5)

DOB	Sex	ZIP	Disease
74	*	941**	aids
74	*	941**	flu
74	*	941**	flu
74	*	941**	aids
64	*	941**	flu
64	*	941**	short breath
64	*	941**	flu
64	*	941**	aids

4-anonymized table



DOB	Sex	ZIP	Disease
74	*	941**	aids
74	*	941**	flu
74	*	941**	flu
74	*	941**	aids

4-anonymized table

An adversary knows that Bob, born in 74 and living in area 94141, is in the table

⇒ Bob belongs to the first group

External knowledge – Example (5)

DOB	Sex	ZIP	Disease
74	*	941**	aids
74	*	941**	flu
74	*	941**	flu
74	*	941**	aids
64	*	941**	flu
64	*	941**	short breath
64	*	941**	flu
64	*	941**	aids

4-anonymized table



DOB	Sex	ZIP	Disease
74	*	941**	aids
74	*	941**	flu
74	*	941**	flu
74	*	941**	aids

4-anonymized table

An adversary knows that Bob, born in 74 and living in area 94141, is in the table

⇒ Bob belongs to the first group

Alice is Bob's wife and she is in the table

External knowledge – Example (5)

DOB	Sex	ZIP	Disease
74	*	941**	aids
74	*	941**	flu
74	*	941**	flu
74	*	941**	aids
64	*	941**	flu
64	*	941**	short breath
64	*	941**	flu
64	*	941**	aids

4-anonymized table



DOB	Sex	ZIP	Disease
74	*	941**	aids
74	*	941**	flu
74	*	941**	flu
74	*	941**	aids

4-anonymized table

An adversary knows that Bob, born in 74 and living in area 94141, is in the table

⇒ Bob belongs to the first group

Alice is Bob's wife and she is in the table

⇒ If Bob has aids, it is highly probable that Alice has aids

Some open issues

- Privacy metrics to model the privacy enjoyed by a data release
- New techniques to protect privacy (in addition to generalization and suppression)
- External knowledge and adversarial attacks
- Novel scenarios of data release, e.g.,
 - release of marginal tables [KG-06]
 - fragmented tables
- Evaluation of privacy vs utility
- Application of k -anonymity-based principles to other scenarios (e.g., social networks, location-based services)